

The Soft Sector: Memory as a Superselection Label and the Harmonic-Number Structure of the Zero-Mode Kernel

Harald Rønning

Abstract. The assembly and interior-net papers built the hard sector of the cone — the generator-diagonal current data and the Haag-Kastler net it determines — and named the excluded cross-generator zero modes as the standing frontier. This note maps that frontier. The soft variable identifies uniquely in this geometry: since the null-projected field $\psi = \lambda\phi$ vanishes at the apex, the per-generator zero mode is pinned at one end, and the soft degree of freedom is the large- λ boundary value $\psi_\infty(\hat{n})$ — which is the memory, $\int_0^\infty \partial_\lambda \psi d\lambda$. Its conjugate, the constant shift, acts trivially on every current: the hard net cannot see it. The central theorem then decides the memory's status: its vacuum variance diverges logarithmically with the smearing window, $V(L) = \frac{1}{2\pi} \ln L + O(1)$ — the slope *universal* — derived in Fourier space and verified numerically (successive-doubling slopes $0.1485 \rightarrow 0.1583$, converging monotonically on $1/2\pi = 0.159155$ with the expected $O(1/L)$ corrections). **The memory is therefore not an operator on the hard sector but a superselection label:** shifted-memory states define disjoint representations, in the classical pattern of infrared representations, and the series' state is the zero-memory sector. The infrared condition thereby acquires its fifth and final duty — after finiteness, necessity, uniqueness, and locality, it is the *choice of superselection sector*. The soft kernel's structure is computed: separated-direction correlations are finite, $1/8\pi^2(1 - c)$, and the Legendre coefficients split as $b_l = \Lambda - 2H_l$ — one universal coincidence divergence Λ (a pure contact term, the entire subtraction ambiguity) plus finite harmonic numbers, via the identity $\int_{-1}^1 (1 - P_l(c))/(1 - c) dc = 2H_l$, verified at 10^{-9} for $l = 1..10$. The label group is supertranslation-shaped — direction-dependent shifts on the sphere — stated as the scalar adjacency of the BMS structure, not its identification. With the interior modular Hamiltonians of the previous paper and the sector map of this one, both prerequisites of the entropic-gravity program are formally in place.

1. The soft variable, identified

The assembly paper computed the undifferentiated cone correlator: off-diagonal in generators, $W_\psi = 1/8\pi^2(1 - c)$, *constant* in the affine parameters — pure zero-mode content, annihilated by the current data $\partial_\lambda \psi$. In this geometry the zero mode has a canonical representative. The null-projected field obeys $\psi(0, \hat{n}) = 0$ at the apex (ϕ regular there), so the per-generator constant is pinned at one end, and the soft degree of freedom is the boundary value at the other:

$$\psi_\infty(\hat{n}) = \lim_{\lambda \rightarrow \infty} \lambda \phi(\lambda(1, \hat{n})) = \int_0^\infty (\partial_\lambda \psi)(\lambda, \hat{n}) d\lambda,$$

which is precisely the *memory*: the total, zero-frequency content of the current along the generator. Its formal conjugate is the constant shift $\psi \mapsto \psi + \sigma(\hat{n})$. The soft sector is this pair, direction by direction.

2. The shift acts trivially on the hard net

Proposition 1. Constant shifts of ψ commute with the entire hard net: every generator of $\mathcal{A}$ is a smeared current, and currents are shift-invariant, $\partial_\lambda(\psi + \sigma) = \partial_\lambda\psi$. The shift therefore defines an automorphism acting as the *identity* on the hard algebra — it is not a symmetry the net represents, but a direction the net cannot see. ■

Whether the conjugate variable — the memory — exists as an operator on the hard sector's Hilbert space is then the entire question, and it is decided quantitatively.

3. Theorem: the memory is a superselection label

****Theorem.**** Let F_L be a smooth window of unit height and length L along a generator (fixed smooth edges), so that $\partial\psi(F_L) \rightarrow \psi_\infty$ formally as $L \rightarrow \infty$. In the assembled (zero-memory) vacuum,

$$V(L) = \langle \partial\psi(F_L)^2 \rangle = \frac{1}{4\pi} \int_0^\infty p |\hat{F}_L(p)|^2 dp = \frac{1}{2\pi} \ln L + O(1),$$

divergent, with universal slope $1/2\pi$. *Derivation:* the chiral vacuum kernel $-1/4\pi(\lambda - \mu - i\epsilon)^2$ has Fourier weight p on the half-line; for the window, $p|\hat{F}_L(p)|^2 \simeq 2/p$ throughout the intermediate region $1/L \ll p \ll 1/(\text{edge})$, contributing $\frac{1}{4\pi} \cdot 2 \ln L$.

Verification: successive-doubling slopes $(V(2L) - V(L))/\ln 2$ for $L = 10 \rightarrow 640$: 0.1485, 0.1536, 0.1563, 0.1577, 0.1583, 0.1583 — monotone on $1/2\pi = 0.159155$ with $O(1/L)$ corrections. ■

Consequences. The limit defining ψ_∞ has no operator realization on the hard GNS space: the memory survives only as a *classical label*. The would-be Weyl operator implementing a memory shift has divergent one-particle cost, so representations built over distinct memory values are disjoint — the classical pattern of infrared (infraparticle) representations, in the tradition of the analyses of Fröhlich, Morchio and Strocchi, and of Buchholz (cited as tradition; no single-theorem attribution intended). The series' state is the zero-memory sector. And with that, the infrared condition of this series receives its fifth and final characterization: having delivered finiteness, been proven necessary, secured uniqueness, and been revealed as locality, it is — at the level of the full cone theory — **the choice of superselection sector**. One condition, five duties, one structural fact throughout: the good variable is the current, and the current's blind spot is a classical label on the world.

4. The kernel: one divergence, then harmonic numbers

****Proposition 2.**** The soft two-point kernel $S(c) = 1/8\pi^2(1 - c)$ is finite at all separated directions and non-integrable at coincidence, and its Legendre content splits as

$$b_l = \int_{-1}^1 \frac{P_l(c)}{1 - c} dc = \Lambda - 2H_l, \quad H_l = \sum_{k=1}^l \frac{1}{k},$$

where $\Lambda = \int dc/(1 - c)$ is one **l-independent** divergence — by completeness, a pure contact term $\propto \delta(c - 1)$ concentrated at coincident directions — while all *l*-dependence is

finite and given by harmonic numbers, via $\int_{-1}^1 (1 - P_l(c))/(1 - c) dc = 2H_l$, verified numerically at 10^{-9} - 10^{-10} for $l = 1, \dots, 10$. ■

Two readings. First, the *subtraction ambiguity* of the soft sector is exactly Λ : a coincident-direction counterterm, invisible in every separated-direction correlation — the physics of the soft kernel at distinct angles is unambiguous. Second, the short-distance behavior $S \sim 2/\theta^2$ makes every coincident-point soft object divergent even after smooth smearing; the soft field is a generalized field whose finite content is its separated-point correlations and its H_l spectrum — the logarithmic mode growth characteristic of boundary Goldstone structures.

5. Adjacency, and the frontier crossed

The label group is the additive group of direction-dependent shifts $\sigma(\hat{n})$ on the sphere — supertranslation-shaped. For this scalar analogue the entire tower $l \geq 0$ consists of memory labels; in the gravitational case the $l = 0, 1$ modes would be momentum and the genuine supertranslations begin at $l \geq 2$ — the spin-2 distinction is noted and not blurred: what is established here is the scalar adjacency of the BMS structure, not its identification. The dynamical point stands regardless: *radiation pumps memory*, so interactions move states between sectors, and the interaction era must treat the sector label as a dynamical variable — the soft sector is where the coupling to the world outside the free net will first show. With the interior modular Hamiltonians of the previous paper and the sector map of this one, the two named prerequisites of the entropic-gravity program — local K 's for $\delta S = \delta\langle K \rangle$, and the home of charges — are formally in place.

6. Scope

Five items. (i) Disjointness of shifted-memory representations is stated in the standard quasi-free pattern (divergent one-particle cost) and attributed to the infrared-representation tradition by name only; no specific-theorem citation is claimed. (ii) The variance ladder's largest windows are quadrature-truncation-limited; the $\ln L$ law and its slope are analytic. (iii) The symplectic pairing of memory and shift is used at the kinematic level; a full charge construction (smeared sphere charges, their algebra) is deferred. (iv) Gravitational memory is spin-2 and outside this note's claims. (v) Contact with celestial-holography structures (the $2/\theta^2$ kernel, the H_l spectrum) is flagged as adjacency for the interaction era.

References

- Companion papers: *The Interior Net: Local Algebras from the Cone and Microcausality as Output; The Assembly: Null Lines Through a Point, the Light-Cone Identification, and Lorentz Symmetry as Output; Bulk Reconstruction: The Interior Two-Point Function from the Cone Data; The Infrared Write-Out: Test Space, Weyl Algebra, and the Verified KMS Structure on the Scale Line; The State Is Unique: Quasi-Free Rigidity, the Closed Zero-Mode Door, and the Infrared Condition's Third Duty.*
- Ashtekar, A. & Streubel, M., on the symplectic geometry of radiative modes at null infinity, *Proc. R. Soc. Lond. A* 376 (1981).

- Strominger, A., *Lectures on the Infrared Structure of Gravity and Gauge Theory*, arXiv:1703.05448.
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Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.